Lecture 3

This lecture is about the postulates of quantum mechanics. You can think of them as the "rules of the game". On a high level, they are about:

1. states
2. time-evolution
3. measurement
4. composite systems

For comparison, we will also look at the corresponding postulates for probabilistic bits. In both cases, I could just state them and you would have to simply accept that that's how it is. But let us try to look more deeply and try to understand their origin (ideally, we would like to derive them from some simple and reasonable assumptions):

1. The state space should be somehow bounded (you don't need infinitely large numbers) and a linear space (if two states are allowed then so are their linear combinations). It contains deterministic states (like 0 and 1).

2. Operations should map states to states and be linear.

3. The probabilities of measurement outcomes sum to one. Each outcome results in a new (deterministic) state.

4. If system A has n deterministic states and B has m then together A and B have $n \times m$ deterministic states.

Probabilistic bits: rules of the game

1. The state space is a linear space over $\mathbb{R}$, consisting of vectors

$$p = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}$$ bounded by $p_i \geq 0$ and $p_1 + p_2 + \dots + p_n = 1$, also known as <u>probability distributions</u>.

2. Transformations are linear and preserve the property of being a probability distribution. That is, if $\sum_{j=1}^{n} p_j = 1$ and $p' = Mp$ then $\sum_{i=1}^{n} p'_i = 1$. Note that

$$p'_i = \sum_{j=1}^{n} M_{ij} p_j \quad \text{so} \quad \sum_{i=1}^{n} p'_i = \sum_{i,j=1}^{n} M_{ij} p_j = \sum_{j=1}^{n} \left(\sum_{i=1}^{n} M_{ij} \right) p_j .$$

We want $\sum_{i=1}^{n} p'_i = 1$ (for all possible p), so we must have $\sum_{i=1}^{n} M_{ij} = 1$. (the j-th column sums to 1)

Similarly, we want $p'_i \geq 0$ which imply $M_{ij} \geq 0$. In other words, each column of M is a probability distribution itself. Such matrices are called <u>stochastic</u>.

3. Measuring p yields outcome i with probability p_i and the state after the measurement is the i-th standard basis vector $\begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \leftarrow i$.

4. If system A is in state i with probability p_i and system B is in state j with probability q_j then together AB is in state (i,j) with probability $p_i q_j$. This is precisely captured by the tensor product:

state of A → $p \otimes q$ ← state of B

$$\begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} \otimes \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \begin{pmatrix} p_1 q_1 \\ p_1 q_2 \\ p_2 q_1 \\ p_2 q_2 \\ p_3 q_1 \\ p_3 q_2 \end{pmatrix} \leftarrow \text{state of } AB$$

Quantum rules of the game

1. The state space is a linear space over $\mathbb{C}$ consisting of vectors $|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}$ such that $|\alpha_1|^2 + |\alpha_2|^2 + \ldots + |\alpha_n|^2 = 1$.

Equivalently, $|\psi\rangle$ is a <u>unit vector</u>: $\langle\psi|\psi\rangle = 1$.

2. Transformations are linear (over $\mathbb{C}$) and map unit vectors to unit vectors. Equivalently, they <u>preserve norms</u>. If $|\psi'\rangle = U|\psi\rangle$ then $\langle\psi'| = \langle\psi|U^\dagger$

$$\langle\psi'|\psi'\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle = 1$$

↑ This has to hold for all states $|\psi\rangle$.

Using an argument similar to one in Lecture 1, the only way this equality can hold simultaneously for all $|\psi\rangle$ is if $U^\dagger U = I$. This means $U^{-1} = U^\dagger$ where U^{-1} denotes the inverse of U. Since $UU^{-1} = I$ too, the conditions

$$U^\dagger U = I \quad \text{and} \quad UU^\dagger = I$$

are equivalent and either of them implies that U is <u>unitary</u>.

3. Measuring $|\psi\rangle \in \mathbb{C}^n$ (in the standard basis) produces outcome $i \in \{1, \ldots, n\}$ with probability $|\langle i|\psi\rangle|^2 = |\alpha_i|^2$ and the state after the measurement is $|i\rangle$.

4. If system A is in state $|A\rangle$ and system B is in state $|B\rangle$ then AB together are in state

$$|A\rangle \otimes |B\rangle.$$

A few words about physics

Quantum mechanics is a branch of physics that describes the behaviour of small systems, such as atoms, photons, and electrons, whose states admit superpositions. It is a framework onto which other physical theories are built upon. For example, quantum field theories such as quantum electrodynamics and chromodynamics. Quantum mechanics itself does not prescribe the state space of particular physical systems, this is done by more specific theories. Any physical system whose state space is $\mathbb{C}^2$ can serve as an implementation of a qubit. For example, the spin of an electron, polarization of a photon, or current in a superconducting circuit. In this course we consider only finite-dimensional systems. But some systems may require an infinite-dimensional Hilbert space. Also, the time evolution of actual physical systems is continuous and described by a differential equation, the Schrödinger equation:

$$i\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$$

whose solution is $|\psi(t)\rangle = U(t)|\psi(0)\rangle$

H: Hamiltonian (a Hermitian matrix)

where $U(t) = \exp(-iHt)$ is unitary.

Expressing a state in a different basis

Any $|\psi\rangle \in \mathbb{C}^n$ can be expressed in the standard basis:

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \sum_{i=1}^{n} \alpha_i |i\rangle = \sum_{i=1}^{n} \langle i|\psi\rangle \, |i\rangle$$

where $\alpha_i = \langle i|\psi\rangle$ is the i-th coordinate of $|\psi\rangle$ (in the standard basis. Similarly, if $|b_1\rangle, \dots, |b_n\rangle$ is any orthonormal basis of $\mathbb{C}^n$ then

$$|\psi\rangle = \sum_{j=1}^{n} \langle b_j|\psi\rangle \, |b_j\rangle$$

where $\langle b_j|\psi\rangle$ is the j-th coordinate of $|\psi\rangle$ in the basis $|b_1\rangle, \dots, |b_n\rangle$. This means that we can express $|\psi\rangle$ in two different ways:

$$|\psi\rangle = \sum_{i=1}^{n} \underbrace{\langle i|\psi\rangle}_{\alpha_i} |i\rangle = \sum_{j=1}^{n} \underbrace{\langle b_j|\psi\rangle}_{\beta_j} |b_j\rangle$$

for some $\alpha_i, \beta_j \in \mathbb{C}$. How are α_i and β_j related? If we left-multiply both sides by $\langle i|$, we get

$$\alpha_i = \sum_{j=1}^{n} \beta_j \underbrace{\langle i|b_j\rangle}_{M_{ij}} = \sum_{j=1}^{n} M_{ij} \beta_j$$

where $M_{ij} = \langle i|b_j\rangle$. In other words, $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = M \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}$ where M is the matrix with entries M_{ij}:

$$M = \sum_{i,j=1}^{n} M_{ij} |i\rangle\langle j| = \sum_{i,j=1}^{n} \langle i|b_j\rangle \cdot |i\rangle\langle j| = \sum_{i=1}^{n} |i\rangle\langle i| \cdot \sum_{j=1}^{n} |b_j\rangle\langle j| = \sum_{j=1}^{n} |b_j\rangle\langle j|.$$

($\sum_{i=1}^{n} |i\rangle\langle i| = I$)

Note that M is unitary since

$$M^\dagger M = \sum_{j,k=1}^{n} |j\rangle\langle b_j|b_k\rangle\langle k| = \sum_{k=1}^{n} |k\rangle\langle k| = I. \quad (\langle b_j|b_k\rangle = \delta_{jk})$$

Thus $M^\dagger |\alpha\rangle = |\beta\rangle$ and

$$\boxed{U = \sum_{j=1}^{n} |j\rangle\langle b_j|}$$

is the basis change from the standard basis $|1\rangle, \dots, |n\rangle$ to the basis $|b_1\rangle, \dots, |b_n\rangle$.

Expressing a matrix in a different basis

We can apply $U = \sum_{i=1}^{n} |i\rangle\langle b_i|$ to go from the standard basis $|i\rangle$ to $|b_i\rangle$. This way we can express a given vector in a different basis. Similarly, given a matrix $A \in M_{n,n}(\mathbb{C})$ with entries $A_{ij} = \langle i|A|j\rangle$, how can we find $B \in M_{n,n}(\mathbb{C})$ such that $B_{ij} = \langle b_i|A|b_j\rangle$?

Note that $B = \sum_{i,j=1}^{n} B_{ij}|i\rangle\langle j| = \sum_{i,j=1}^{n} |i\rangle\langle b_i|A|b_j\rangle\langle j| = UAU^{\dagger}$, where U is the same basis change unitary as before.

Measurement in a different basis

More generally, we can measure a state $|\psi\rangle \in \mathbb{C}^n$ in any orthonormal basis $|b_1\rangle, \dots, |b_n\rangle$ of $\mathbb{C}^n$. Such measurement produces outcome $i \in \{1, \dots, n\}$ with probability $|\langle b_i|\psi\rangle|^2$ and the corresponding post-measurement state is $|b_i\rangle$. Note that this is equivalent to applying $U = \sum_{i=1}^{n} |i\rangle\langle b_i|$, performing the standard basis measurement and then applying $U^{\dagger}$. This is called an orthogonal measurement.

Example: If you measure

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \text{ and } |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

in the standard basis, you get each of the two outcomes with probability $\frac{1}{2}$, so for this measurement the two states appear indistinguishable. Of course, this is not the best measurement to try! You should instead map them to the standard basis by applying

$$H = |0\rangle\langle+| + |1\rangle\langle-| = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} + \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

and then measure. This way you are simply trying to discriminate between $|0\rangle$ and $|1\rangle$, which you can do perfectly. Alternatively, since $|+\rangle$ and $|-\rangle$ is an orthonormal basis, you can directly measure in this basis to perfectly discriminate these states. More generally, you can perfectly discriminate any orthonormal set of states by simply measuring in the corresponding basis.

Tensor product and the computational basis

Given a qubit in state $|0\rangle$ and another one in state $|1\rangle$, their joint state is $|0\rangle \otimes |1\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}$.

We abbreviate this simply as $|0,1\rangle$ or $|01\rangle$.

Similarly: $|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}$ $|01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}$ $|10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}$ $|11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}$

where the binary encoding of the location (starting from 0) of the entry 1 always coincides with the string inside the ket.

These are called the computational basis states.

They represent different bit strings that can be stored in a quantum computer. A general two-qubit state is a superposition of these four states:

$$\alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

$$= \begin{pmatrix}\alpha_{00}\\ \alpha_{01}\\ \alpha_{10}\\ \alpha_{11}\end{pmatrix} \begin{matrix}|00\rangle\\ |01\rangle\\ |10\rangle\\ |11\rangle\end{matrix}$$

The order of the computational basis states is always alphabetical (or increasing for binary strings).

Similarly, $|111\rangle \in \mathbb{C}^8$ is a vector with 1 in the last (7-th when starting from 0) component.

Correlated versus product distributions

Given two probabilistic bits $p=\begin{pmatrix}p_0\\p_1\end{pmatrix}$ and $q=\begin{pmatrix}q_0\\q_1\end{pmatrix}$, their joint distribution is

$$p\otimes q=\begin{pmatrix}p_0\\p_1\end{pmatrix}\otimes\begin{pmatrix}q_0\\q_1\end{pmatrix}=\begin{pmatrix}p_0\begin{pmatrix}q_0\\q_1\end{pmatrix}\\p_1\begin{pmatrix}q_0\\q_1\end{pmatrix}\end{pmatrix}=\begin{pmatrix}p_0q_0\\p_0q_1\\p_1q_0\\p_1q_1\end{pmatrix}\begin{matrix}00\\01\\10\\11\end{matrix}$$

Such distributions are called **product**. They represent independent random bits. But is every distribution a product distribution? That is, for a given $r=\begin{pmatrix}r_{00}\\r_{01}\\r_{10}\\r_{11}\end{pmatrix}$, can we always find $p=\begin{pmatrix}p_0\\p_1\end{pmatrix}$ and $q=\begin{pmatrix}q_0\\q_1\end{pmatrix}$ such that $r=p\otimes q$?

Example If $r=\frac{1}{2}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}=\begin{pmatrix}p_0q_0\\p_0q_1\\p_1q_0\\p_1q_1\end{pmatrix}$ then $p_0q_1=0$, implying $p_0=0$ or $q_1=0$. The first case contradicts $p_0q_0=\frac{1}{2}$ and the second case contradicts $p_1q_1=\frac{1}{2}$, implying that r is **not** a product distribution.

Distributions that are not product are called **correlated**.

Product and entangled states

Similar terminology applies also to two-qubit quantum states. If $|\Psi\rangle \in \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$ is of the form $|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$ for some $|\psi_1\rangle, |\psi_2\rangle \in \mathbb{C}^2$, it is called a product state. Otherwise it is called entangled.

Example $\frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
so it is a product state.

Example $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ are both entangled.

Note: Physical separation does not imply that the joint state must be product. Just like two distant random coins can still be correlated, two physically separated particles can still be entangled, as long as they had a chance to interact at some point in the past.

Measuring one out of two bits

Given two probabilistic bits in state $\begin{pmatrix} r_{00} \\ r_{01} \\ r_{10} \\ r_{11} \end{pmatrix}$ the probability of observing the first bit in state 0 is $r_{00} + r_{01}$. The state of the remaining bit is

$$\begin{cases} 0 & \text{w.p. } \dfrac{r_{00}}{r_{00}+r_{01}} \\ 1 & \text{w.p. } \dfrac{r_{01}}{r_{00}+r_{01}} \end{cases}$$

In particular, if the two bits are in a product state then $r_{00} = p_0 q_0$ and $r_{01} = p_0 q_1$ so the second bit q of $p \otimes q$ is not affected if the first bit is measured since

$$\frac{r_{00}}{r_{00}+r_{01}} = \frac{p_0 q_0}{p_0(q_0+q_1)} = \frac{q_0}{q_0+q_1} = q_0 \text{ since } q_0+q_1 = 1$$

and $$\frac{r_{01}}{r_{00}+r_{01}} = \frac{p_0 q_1}{p_0(q_0+q_1)} = \frac{q_1}{q_0+q_1} = q_1.$$

On the other hand, if the two bits are correlated the state of the unmeasured bit depends on the outcome of the measurement performed on the other bit.

Example If the first bit of $\frac{1}{2}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ is measured, each outcome occurs with probability 1/2 and the state of the second bit is equal to the measurement outcome from the first bit.

Measuring one out of two qubits

The rule for this is similar, except we are dealing with amplitudes instead of probabilities. Let's say we measure the first qubit of $\psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle$.

We get outcome $\begin{cases} 0 & \text{w.p. } |\psi_{00}|^2 + |\psi_{01}|^2 = p_0 \\ 1 & \text{w.p. } |\psi_{10}|^2 + |\psi_{11}|^2 = p_1 \end{cases}$

If the outcome was $k \in \{0,1\}$, the state on the second qubit is

$$\frac{1}{\sqrt{p_k}}\left(\psi_{k0}|0\rangle + \psi_{k1}|1\rangle\right)$$

This is a unit vector since $p_k = |\psi_{k0}|^2 + |\psi_{k1}|^2$.

Just like with probabilistic bits, the state of the remaining qubit is unaffected by a measurement of the first qubit if the initial state was a product state. Otherwise (if the qubits were entangled), the second qubit's state depends on the outcome of the measurement on the first qubit.